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A novel fruit fly optimization algorithm with Levi flight and challenge probability

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Abstract

Aiming at the problem of low convergence precision and premature convergence of fruit fly optimization algorithm (FOA), this paper proposes a novel fruit fly optimization algorithm with Levi flight and challenge probability (LCFOA). We introduce the Levi flight mechanism and challenge adjustment process to coordinate the global exploration and local exploitation capabilities of the algorithm. Then, we designed an adaptive challenge factor in the challenge adjustment process to improve the convergence precision of the algorithm. The convergence performance of the algorithm is verified by 6 different benchmark functions. The numerical experimental results confirmed that the proposed LCFOA has better search precision and convergence speed.

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1. Introduction

Most of the problems in the field of engineering science can be summarized as optimization problems. Swarm intelligence algorithm is a heuristic algorithm based on the idea of evolution, which uses the mutual cooperation and information sharing between organisms to perform global optimization. Among them, FOA¹ has been applied to deal with scheduling problems^{2,3} and parameter setting problems⁴ due to its characteristics such as easy implementation, few adjustable parameters and good local search ability. However, with the popularization of the

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algorithm, its disadvantages such as premature convergence and weak global search ability are gradually revealed. For this reason, Wang⁵ proposed an improved fruit fly optimization algorithm based on joint search strategy, which has a high search accuracy in solving high-dimensional problems. Pan⁶ introduced a complex adaptive parameter to adjust the search radius of fruit flies, and the algorithm's local search capability was enhanced. However, due to the specific candidate solution generation mechanism of FOA, the optimization performance of the algorithm is highly dependent on the current optimal solution, which restricts its ability to solve complex optimization problems.

The primary focus of this paper is to improve the global optimization ability of FOA and reduce the probability of premature convergence and propose a fruit fly optimization algorithm with Levi flight and challenge probability (LCFOA). The proposed algorithm introduces levy flight to update the step size in osphresis foraging, uses a challenge probability in vision foraging to adjust the current candidate solution. Finally, we verify the convergence performance of the LCFOA through 6 benchmark functions.

The rest of this paper is outlined as follows. Section 2 presents the pure FOA while LCFOA is detailed in Section 3. To verify the superior performance of LCFOA, some experimental results are given in Section 4. Finally, Section 5 offers the conclusion of the paper.

2. Original fruit fly optimization algorithm

Fruit flies have keen olfactory and visual organs, allowing them to quickly capture the smell of food in the air and detect and fly to the position of their companions closer to the food. Inspired by this, FOA can be described as the following steps.

Step 1: Initialization parameter. Set the maximum number of iterations as $G_{\max}$, the fruit fly swarm size as N . $X_{i,G}$, $Y_{i,G}$ is the optimal position of current G^{th} generation and $X_{b,0}$ and $Y_{b,0}$ are randomly set within the range of $[0, 1]$.

Step 2: Osphresis foraging. Fruit flies use its sensitive sense of smell to search for odors in the air. R is a random number within the range of $[-1, 1]$.

$$\begin{aligned} X_{i,G} &= X_{b,G-1} + R, i = 1, 2, \dots, n \\ Y_{i,G} &= Y_{b,G-1} + R, i = 1, 2, \dots, n \end{aligned} \quad (1)$$

Step 3: Calculate the distance $D_{i,G}$ between fruit fly and the origin, and then calculate the smell concentration judgment value $S_{i,G}$.

$$D_{i,G} = \sqrt{X_{i,G}^2 + Y_{i,G}^2}, S_{i,G} = \frac{1}{D_{i,G}} \quad (2)$$

Step 4: Calculate the smell concentration of each fly.

$$F_{i,G} = \text{Function}(S_{i,G}) \quad (3)$$

Step 5: Preserve the information of the optimal smell concentration in the population:

$$[bestSmell \ bestindex] = \min(F_{i,G}) \quad (4)$$

Step 6: Vision foraging. The fruit fly swarm use its sensitive vision to find the specific location of food and fly to it. This process records the best smell concentration $Smell_G$ and its location information:

$$Smell_G = bestSmell, X_{b,G} = X_{bestindex,G}, Y_{b,G} = Y_{bestindex,G} \quad (5)$$

Step 7: Begin the iteration optimization, repeat steps 2-6 to calculate whether the smell concentration is lower than the previous one until the preset G_{max} or precision is achieved.

3. A modified fruit fly optimization algorithm

As a general algorithm, FOA is simple in structure and easy to implement, but it often falls into local optimization. The flight direction and distance of osphresis foraging are irregular, and blind flight will reduce the search efficiency of the algorithm. Therefore, we introduce Levi flight into osphresis foraging to adjust the direction and distance of FOA. Levi flight is a kind of walking mode between short distance search and occasionally long distance walk. Therefore, it can increase the population diversity and expand the search range, making the optimization algorithm has the ability to jump out of the local optimal and reduce the probability of premature convergence. In addition, a challenge probability P_a is introduced in vision foraging to challenge the optimal position, which is designed to improve the search precision of FOA. The specific steps of LCFOA can be described as follows:

Step 1: Initialization parameter. Set the maximum number of iterations as G_{max} , the fruit fly swarm size as N , and set the location of fruit fly $X_{i,0}$, $Y_{i,0}$ and optimal location $X_{b,0}$, $Y_{b,0}$ randomly within the range of $[0, 1]$.

Step 2: Update the position of the fruit fly through Levi flight⁷:

$$\begin{aligned} X_{i,G} &= X_{b,G-1} \pm (X_{i,G-1} - X_{b,G-1}) \otimes L(\beta), \quad i = 1, 2, \dots, n \\ Y_{i,G} &= Y_{b,G-1} \pm (Y_{i,G-1} - Y_{b,G-1}) \otimes L(\beta), \quad i = 1, 2, \dots, n \end{aligned} \quad (6)$$

Step 3: Implement FOA's steps 3-5;

Step 4: Generate a random number P_t , when $P_t < P_a$, go to step 5, otherwise go to step 6 of FOA.

Step 5: Take the difference between the optimal solution $X_{bestindex,G}$, $Y_{bestindex,G}$ and the worst solution $X_{worstindex,G}$ and $Y_{worstindex,G}$ of the population as the challenge factor to challenge the current optimal solution.

$$\begin{aligned} [worstSmell \text{ worstindex}] &= \max(F_{i,G}) \\ X_{b,G} &= X_{bestindex,G} - \frac{G_{max}}{G} \log\left(\frac{G}{G_{max}}\right) (X_{bestindex,G} - X_{worstindex,G}) \\ Y_{b,G} &= Y_{bestindex,G} - \frac{G_{max}}{G} \log\left(\frac{G}{G_{max}}\right) (Y_{bestindex,G} - Y_{worstindex,G}) \end{aligned} \quad (7)$$

Step 6: Enter optimization and repeat steps 2-4 to determine whether the smell concentration is better than before. The iteration terminates when a predefined G_{max} or precision is reached.

4. Experiments

In this section, we use benchmark functions to test the convergence performance of pure FOA and the proposed LCFOA. We select four common benchmark functions, including two unimodal functions and two multi-peak functions, for comparison experiments, as shown in Table 1, Where Dim is the dimension of the benchmark function.

In addition, in order to further verify the ability of LCFOA to solve complex optimization problems, on the basis of the above functions, two combined multi-peak functions are given in Table 2. In general, the functions in Table 2 are more complex than those in Table 1.

Table 1. Benchmark functions(“U” and “M” represent unimodal and multi-model respectively)

Function	Equation	Boundary	Optima	Peak
f_1	$f_1 = \sum_{i=1}^{Dim} x_i^2$	$[-100,100]$	0	U
f_2	$f_2 = \sum_{i=1}^{Dim} x_i^2 + (\sum_{i=1}^{Dim} 0.5ix_i)^2 + (\sum_{i=1}^{Dim} 0.5ix_i)^4$	$[-10,10]$	0	U
f_3	$f_3 = 20 + e - 20e^{-\frac{1}{5}\sqrt{\frac{1}{n}\sum_{i=1}^n x_i^2}} - e^{-\frac{1}{n}\sum_{i=1}^n \cos(2\pi x_i)}$	$[-32,32]$	0	M
f_4	$f_4 = 10Dim + \sum_{i=1}^{Dim} [x_i^2 - 100\cos(2\pi x_i)]$	$[-5.12,5.12]$	0	M

Table 2. Composition functions(“M” represent multi-model function)

Function	Equation	Boundary	Optima	Peak
f_5	$f_5 = f_1 + 0.25f_3$	$[-32,32]$	0	M
f_6	$f_6 = f_2 + 0.75f_3$	$[-32,32]$	0	M

Two different experiments were design to compare the convergence precision and convergence speed of FOA and LCFOA. The experimental results adopt the average value of each algorithm independently run 50 times

The initial parameters of the algorithm are set as follows: the fruit fly swarmsize $N=15$;Themaximum number of iterations $G_{\max}=500$;The initial positions $X_{b,0}$, $Y_{b,0}$, $X_{i,0}$ and $Y_{i,0}$ are randomly generated within the range of $[0, 1]$.The simulation environment was programmed in Matlab R2014b under the Windows 10 environment, and the numerical experiment was carried out on the Inter(R) Xeon(R) CPU E5-2687W V3@3.10 GH 10-core processor, RAM 64GB memory.

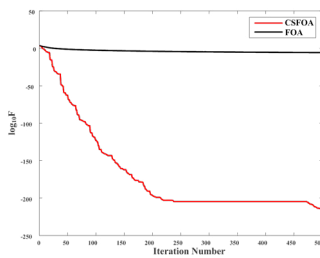
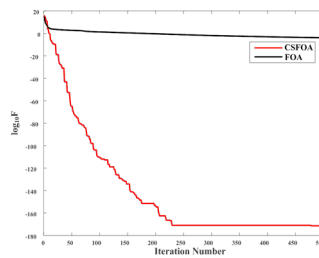
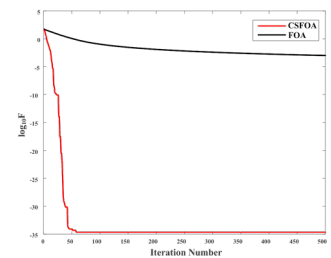
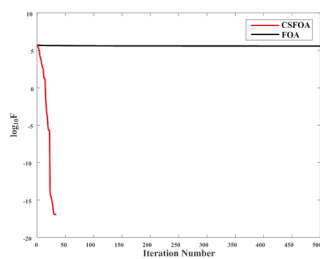
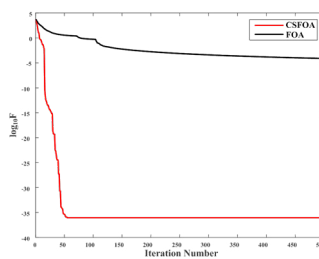
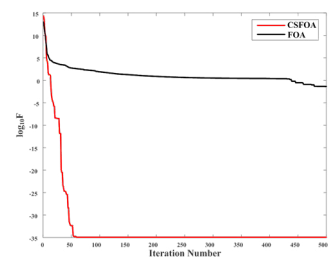
4.1. Comparison of convergence precision and convergence speed between LCFOA and FOA

The main purpose of our first experiment is to compare the search precision and convergence speed of FOA and the proposed LCFOA under a specified number of iterations. The comparison results of convergence precision are shown in Table 3, which Mean stand for the average value of the optimal solution in 50 repeated experiments, and Std represents the standard deviation. The smaller the Mean and Std are, the better.

Looking at Table 3, it is apparent that the Mean and Std of LCFOA are superior to FOA in all benchmark functions. For benchmark functions $f_3 \sim f_6$, the Std of LCFOA is 0, which fully demonstrates the optimization stability of LCFOA. In particular, Rastrigin f_4 is a complex multi-peak function that has a large number of local minima but only one global minimum, making the optimization algorithm easy to fall into its local minima. However, LCFOA still skips its local minimum and quickly gets the theoretical global minimum.

Table 3. Performances comparison of FOA and LCFOA

Function	Dim	LCFOA		FOA	
		Mean	Std	Mean	Std
f_1	10	1.37E-146	9.71E-146	3.98E-04	1.15E-05
	20	7.50E-111	5.30E-110	1.60E-03	4.68E-05
	30	1.27E-94	8.99E-94	3.60E-03	1.10E-04
f_2	10	1.09E-123	7.70E-123	4.87E-04	6.94E-05
	20	4.23E-84	2.99E-83	3.17E-03	4.28E-04
	30	1.55E-75	1.10E-74	2.21E-02	1.07E-02
f_3	10	8.88E-16	0	1.14E-01	6.12E-01
	20	8.88E-16	0	1.30E-01	6.38E-01
	30	8.88E-16	0	5.03E-02	1.30E-03
f_4	10	0	0	5.26E+01	6.85E+00
	20	0	0	1.54E+02	1.09E+01
	30	0	0	2.66E+02	1.95E+01
f_5	10	2.22E-16	0	1.29E-01	8.61E-01
	20	2.22E-16	0	1.17E-02	7.00E-04
	30	2.22E-16	0	1.63E-02	9.32E-04
f_6	10	6.66E-16	0	2.95E-01	1.93E+00
	20	6.66E-16	0	3.96E-02	3.27E-03
	30	6.66E-16	0	2.57E-01	1.24E+00

(a) f_1 (b) f_2 (c) f_3 (d) f_4 (e) f_5 (g) f_6 Fig. 1. Fitness evolutionary curves of $f_1 \sim f_6$.

The fitness value evolution curve of LCFOA and FOA on the six benchmark functions is shown in Figure 1, where the abscissa is iteration number and the ordinate represents the logarithm of the average fitness value in the 50 repeated experiments.

Figure 1 shows that as iteration number increases, the fitness value of each iteration in FOA tends to converge very slowly, while LCFOA always keeps a downward trend. In terms of convergence speed, the proposed LCFOA is faster than FOA. Figure 1 also shows that during algorithm evolution, FOA is in a stagnant state because it is trapped in local optima. But LCFOA is capable to jump out of local minima and evolve until it obtains a global optimal value or reaches a termination condition.

4.2. Comparison of the convergence success rate and the number of iterations under preset precision

Secondly, we compare the two algorithms by the convergence success rate and iteration number under a preset precision. Table 4 shows the numerical experimental results, where Mean, Min and Max respectively represent the average number, the minimum number and the maximum number of iterations that meets the preset precision requirements in 50 repeated experiments. Success Rate represents the proportion of the number of experiments that successfully converge to the preset precision in 50 experiments. " ∞ " indicates that the convergence precision is still less than the preset precision when the iteration number reaches the maximum 500. The smaller Mean, Min and Max and the greater the success rate, the better.

Table 4. Comparison of the convergence success rate and the number of iterations under preset precision

Function(Preset Precision)	$f_1(10^{-1})$		$f_2(10^{-5})$		$f_3(10^{-1})$		$f_4(10^{-5})$		$f_5(10^{-15})$		$f_6(10^{-15})$	
Algorithm	LCFOA	FOA	LCFOA	FOA	LCFOA	FOA	LCFOA	FOA	LCFOA	FOA	LCFOA	FOA
Mean	5.72	91.82	9.56	∞	6.08	280.36	11.54	∞	28.36	∞	31.18	∞
Min	2	81	2	∞	2	257	3	∞	9	∞	7	∞
Max	22	102	31	∞	17	434	43	∞	58	∞	55	∞
Success Rate	1	1	1	0	1	1	1	0	1	0	1	0

From the data in the above table, compared with LCFOA, except for f_1 and f_3 , FOA fails to converge to the preset precision in almost all benchmark functions, while LCFOA meets the preset precision in all experiments. According to Table 4, even for functions f_1 and f_3 , both algorithms can converge to the specified precision, but the required Mean, Min and Max of FOA are 4 ~ 128 times of LCFOA.

These experiments confirm that the proposed LCFOA is superior to the basic FOA in terms of convergence speed and convergence precision, that is, the convergence performance of LCFOA is superior to the basic FOA.

5. Conclusion

In this paper, we propose a novel fruit fly optimization algorithm called LCFOA. Four benchmark functions and two complex combined benchmark functions are used to verify the convergence performance of the algorithm. The numerical experimental results indicate that LCFOA improves greatly in convergence speed and precision. Future work will focus on testing the performance of LCFOA in engineering applications.

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